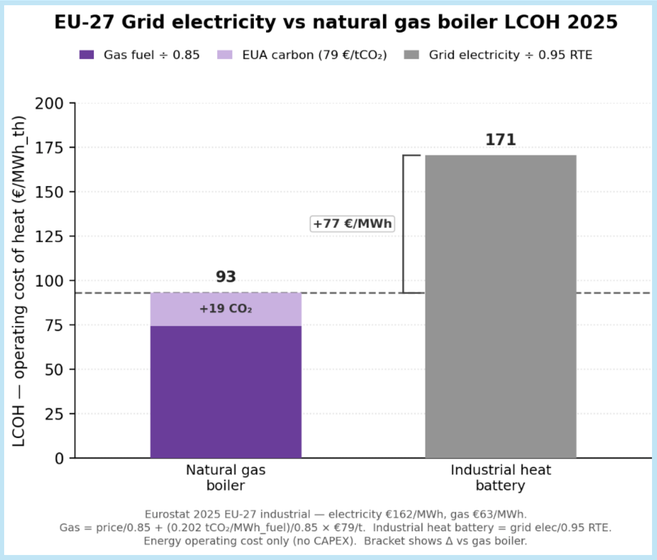
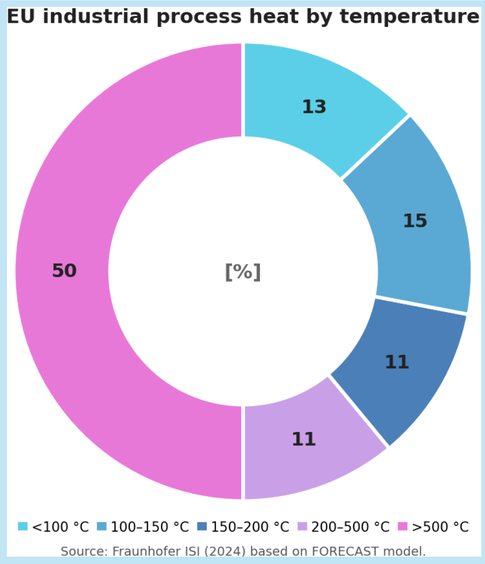
Techno-economic Potential of Electrified Industrial Process Heat in European Manufacturing

Anna Yuen[1, 2, *], Amol Phadke[2]

[1]Energy and Resources Group, University of California, Berkeley
[2]Goldman School of Public Policy, University of California, Berkeley

Introduction





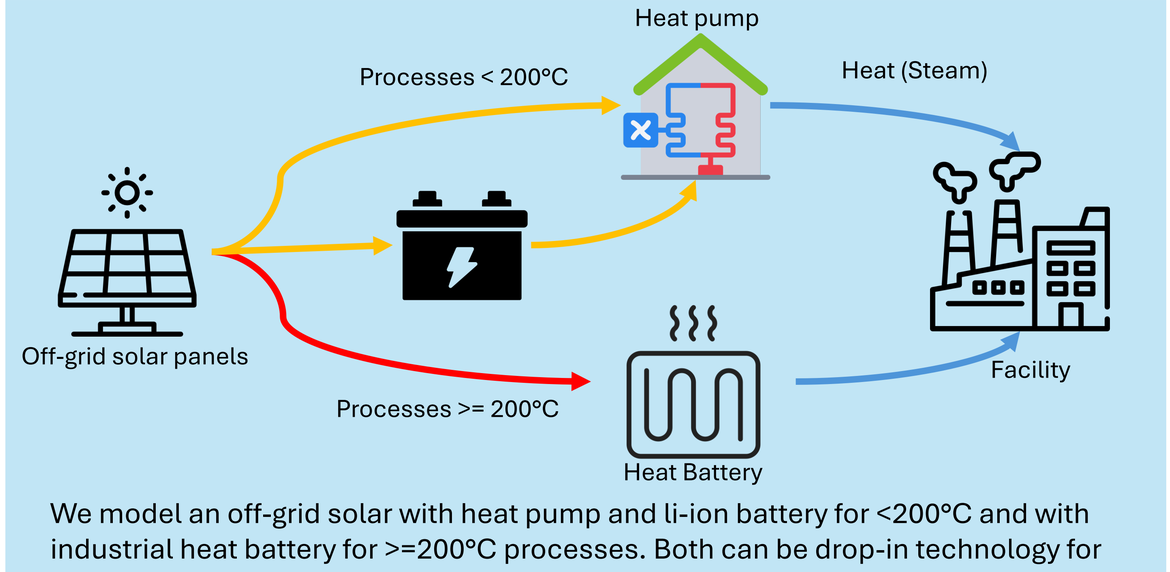
50% of EU industrial process heat demand is above 500°C.
Industrial heat batteries are the leading option to electrify it, yet grid power costs €77/MWh more than gas.

Results

	5 km	15 km
Served (max heat)	58.8% (307 TWh)	84.4% (441 TWh)
LCOH (max heat)	€55	€62
Served (max savings)	**53.4% (278 TWh)**	**73.9% (377 TWh)**
LCOH (max savings)	**€41**	**€42**

Max heat: for each site, raise availability as high as possible while keeping LCOH below that country's gas heat cost. Goal: **serve the most heat that still beats gas**.
Max savings: for each site, pick the availability that maximizes total € savings vs gas:
served heat × (gas cost − LCOH). Goal: **largest bill savings**, even if that means serving less heat.

Methods



We model an off-grid solar with heat pump and li-ion battery for <200°C and with industrial heat battery for >=200°C processes. Both can be drop-in technology for steam generation in manufacturing facilities.

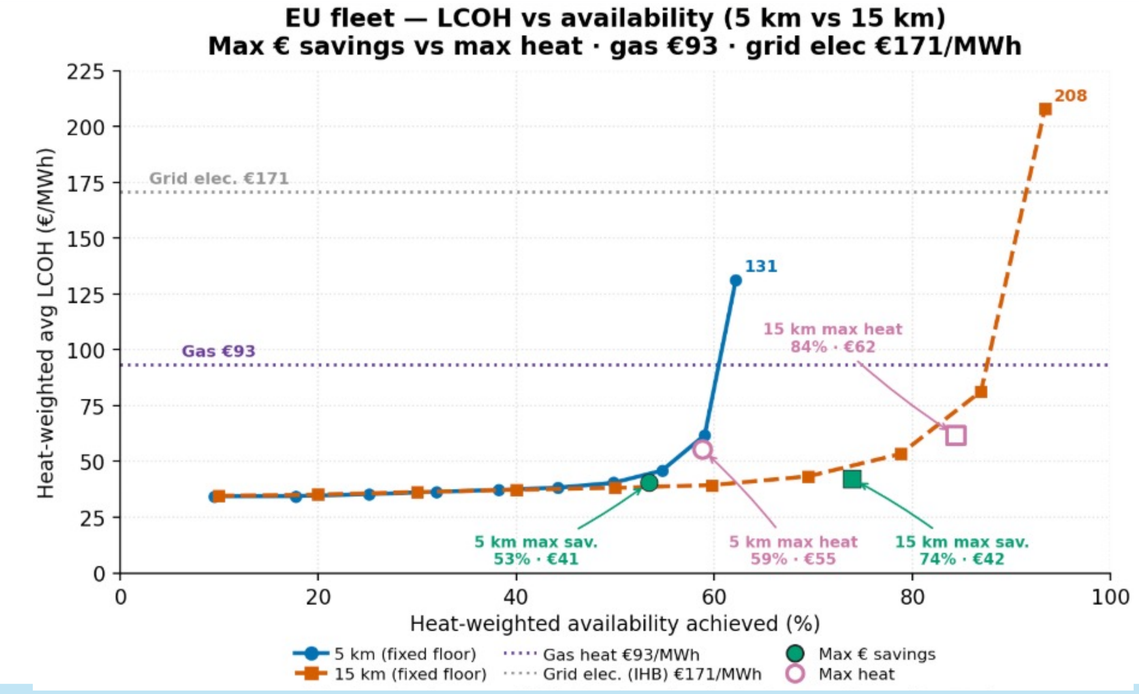
Facility-level Production Data | **Solar Radiation** | **Land Availability**

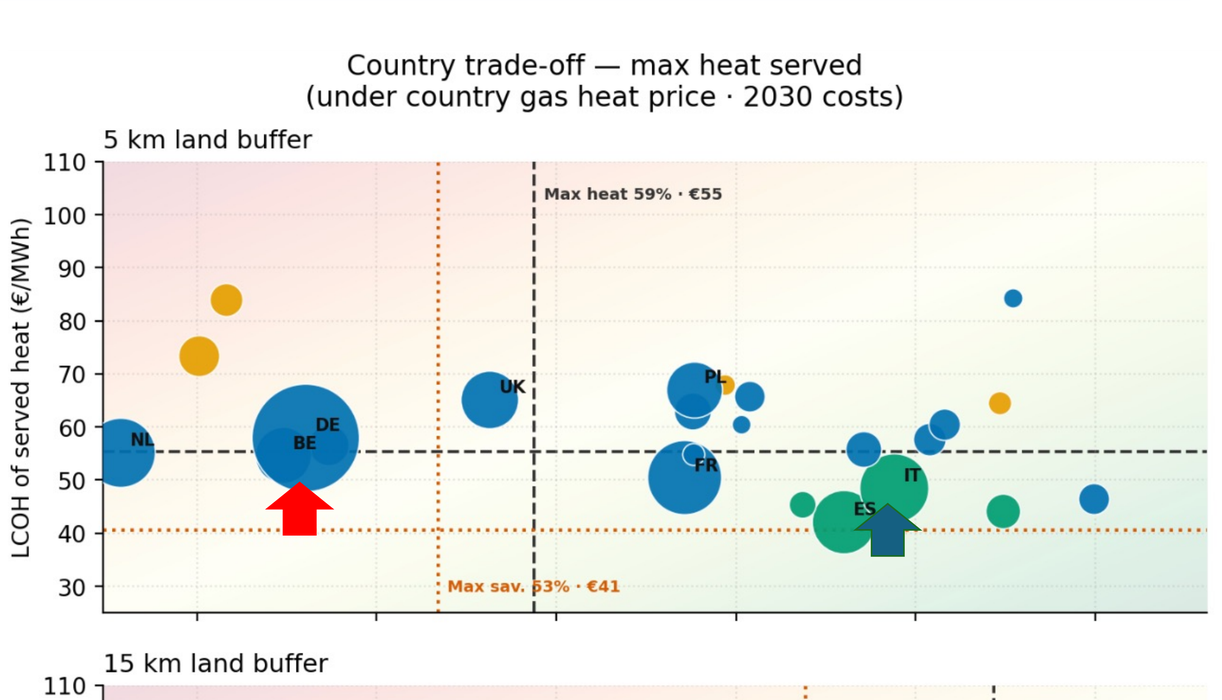
Optimization
Minimize LCOH subject to energy balance, solar capacity and land availability

% of total heat decarbonized and savings

With the total heat level decarbonized at 10-100% availability, we analyzed the costs and benefit of implementing solutions at different availability served for different sectors and countries.

Discussion





To maximize savings, a facility should aim for availability up to the maximum savings points. The remaining can be met with grid electricity or gas.
To maximize heat demand served, a facility could overbuild solar panels while still being under gas prices resulting in a lower marginal benefit but is still gas competitive.

- Steam cracking, iron & steel sector have very large sites and have more heat intense processes (> 500°C) resulting in much higher heat loads.
- These sectors require more hybrid solutions: increased land availability, utilizing grid electricity, co-firing with gas plants.
- Even building up to the max savings point, solar plus storage can meet over 50% of the heat served.

Conclusion

- Germany, the Netherlands, and France have very large process-heat demand. With a ~15 km land buffer, a large majority can be served below country gas. At 5 km, Germany and the Netherlands remain more land-constrained.
- Southern Europe (Greece, Spain, Italy, Portugal) is the strongest fit: ~95% of demand served at LCOH roughly **70–75% of country gas** (closer to ~half gas under a max-savings framing). Sites tend to be smaller, with better solar yield.
- For high-temperature processes, this solution can decarbonize a high share of **replaceable** heat (~80% for steel and steam cracking at 15 km), even if not 100% everywhere.
- By meeting heat on-site with solar + storage, this package can reduce reliance on gas boilers and incremental electric load on the local grid, potentially easing upgrades and gas exposure, and offers a deployable path to cheaper cleaner heat.